Internship Report

Normal Vector estimation

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Introduction

Estimating normal vectors from 3D point clouds is a fundamental task in computer vision, computer graphics, and 3D data processing. Despite its importance, the problem remains challenging due to the presence of noise, outliers, non-uniform sampling densities, and sharp geometric features. Traditional approaches, such as Principal Component Analysis (PCA)-based methods, often struggle to deliver accurate results under these conditions. To overcome these limitations, recent research has introduced a variety of improved techniques. Some approaches extend classical geometry-based methods by integrating robust statistical measures, while others adopt learning-based strategies, including neural networks and graph neural networks, to adaptively estimate normals. For example, DeepFit combines neural networks with weighted least squares to improve robustness, while HSurf-Net employs implicit hypersurface representations to predict normals directly. Collectively, these works reflect the field's transition from purely geometric formulations toward more flexible, data-driven solutions.

This report presents the work carried out during the internship, which followed a process spanning literature review, implementation, testing, and deployment. Specifically, the approach adopted during the internship was based on eigenvalue-driven region growing combined with angle-based merging of regions to estimate normals. This method was explored as a practical and effective alternative to both traditional PCA and fully learning-based strategies, balancing geometric interpretability with computational efficiency. The report details the methodology, experimental evaluations, and the integration of this solution into a usable system.

Literature review

Normal vector estimation from 3D point clouds has been widely studied, with approaches evolving from classical geometry-based methods to modern learning-based models. Traditional techniques, such as PCA, are simple and computationally efficient but suffer in the presence of noise, outliers, non-uniform sampling, and sharp surface features. To address these limitations, several improved methods have been proposed.

Learning-based methods have shown significant progress. *DeepFit* [2] combines neural networks with weighted least squares, achieving robustness against noise and outliers while enabling extraction of normals and other geometric properties. Similarly, *Deep Iterative Surface Normal Estimation* [3] uses graph neural networks to iteratively learn adaptive anisotropic kernels, which are then used in weighted least-squares plane fitting. This method is particularly effective on raw point cloud data, requiring no preprocessing. Another learning-based approach, *HSurf-Net* [5], introduces implicit hypersurface learning with multi-layer perceptrons, directly predicting normals without explicit surface fitting, and demonstrating strong performance on both synthetic and real-world datasets.

Geometry-based improvements also exist. In *Robust and Edge-Aware Normal Vector Estimation* [4], the authors extend PCA by incorporating robust statistics and feature awareness. By first identifying local neighborhoods and then applying a robust PCA that automatically down-weights outliers, this method achieves better reliability in both smooth and feature-rich regions.

Across these works, key challenges such as robustness to noise, handling of outliers, adaptation to non-uniform sampling, and preservation of sharp features are addressed in different ways. Geometry-based approaches emphasize

efficiency and interpretability, while learning-based methods provide adaptability and higher accuracy at the cost of requiring training data and more computation.

Overall, the field is moving toward data-driven solutions that can generalize across diverse conditions without heavy preprocessing. However, classical methods remain relevant due to their efficiency and ease of implementation. Combining the strengths of both — feature-awareness, robustness, and adaptability — represents a promising direction for future work.

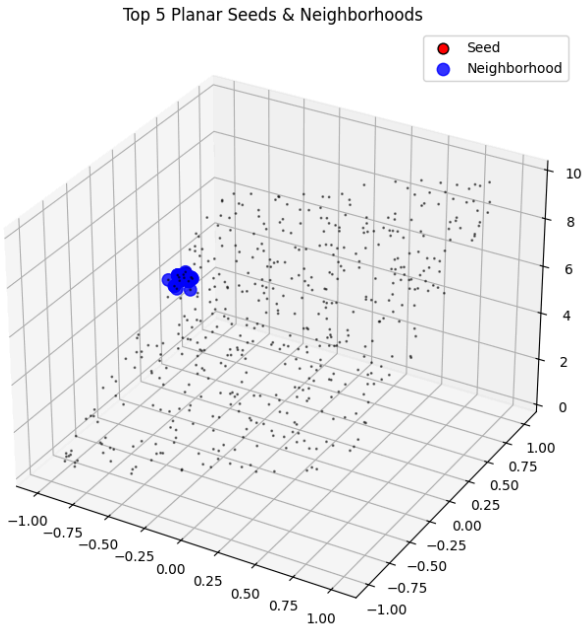
Method Explanation

The proposed method for normal vector estimation is based on **region-growing segmentation combined with eigenvalue analysis**. It consists of three main stages: seeding, planar segmentation, and merging.

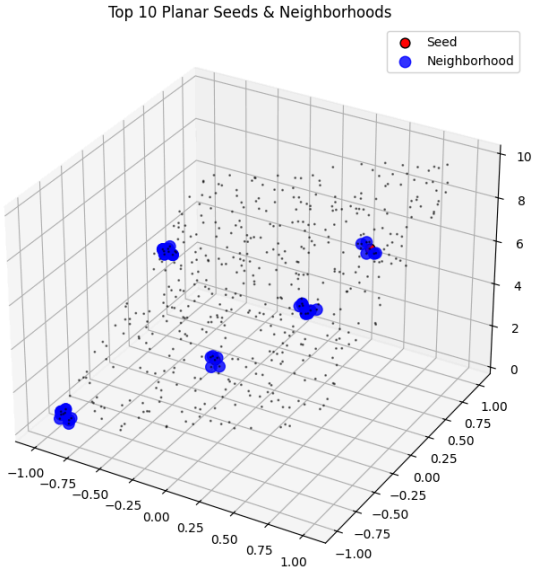
1. Seeding Strategy

To initialize region growing, different seeding techniques were explored:

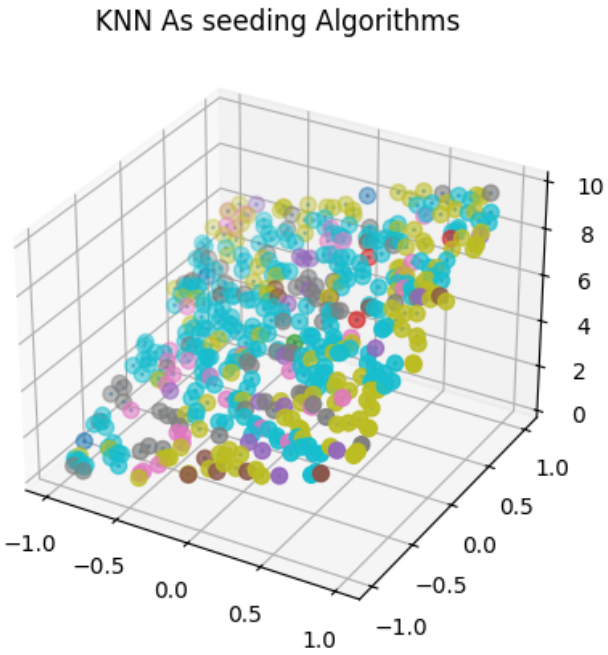
- **Highest Local Planarity**: Points with the highest local planarity score are chosen as seeds. The score is computed from the eigenvalue ratios of the covariance matrix of each point's neighbors, ensuring that highly planar regions are prioritized.



- **Density-Based Seeding**: Seeds are selected in areas with high local point density, which improves coverage across the point cloud.



- **k-Nearest Neighbors (k-NN) Seeding**: Each point selects its neighborhood based on a fixed number of nearest neighbors, ensuring consistent neighborhood sizes for the covariance estimation.



2. Planar Segmentation (Region Growing)

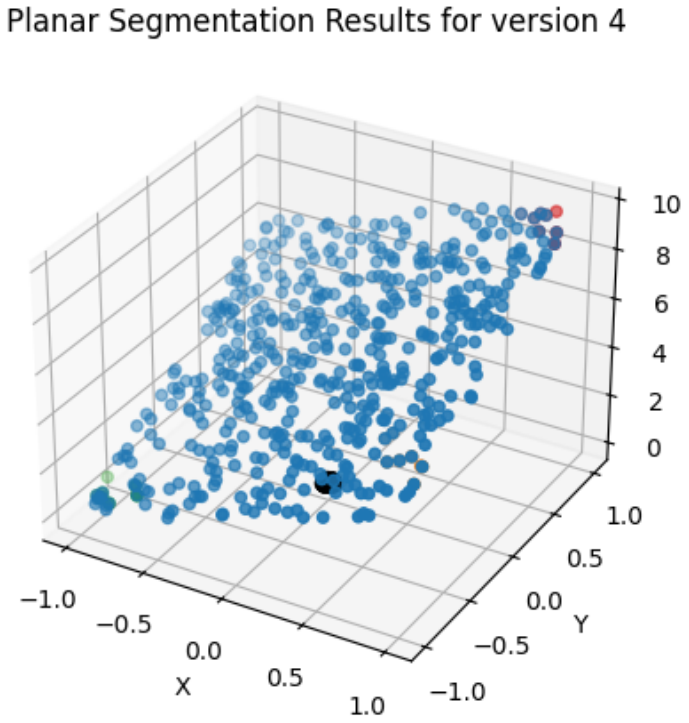
Starting from the seeds, neighboring points are incrementally added to form planar regions. A point is included if it satisfies an **eigenvalue ratio condition**: the smallest eigenvalue of the covariance matrix of the extended region must not deviate significantly from the seed's initial eigenvalue. This criterion is controlled by a user-defined threshold.

At each iteration:

- The covariance matrix of the current region is recomputed.
- Eigen decomposition provides both the **normal vector** (corresponding to the smallest eigenvalue) and a **planarity score**, defined as:

$$\text{Planarity Score} = \frac{\gamma 2 - \gamma 1}{\gamma 3}$$

where $\lambda 1 \leq \lambda 2 \leq \lambda 3$ are the eigenvalues of the covariance matrix.



3. Merging Step

After planar segmentation, multiple small planar patches may be identified. To avoid redundancy, regions with **similar normal vectors** are merged. The similarity between two normals is evaluated using the angle between them:

- If the angle between two regions' normals is smaller than a predefined threshold, the regions are merged into a larger planar surface.
- This step reduces fragmentation and ensures that each detected plane corresponds to a meaningful surface.

4. Final Output

The method outputs a set of merged planar regions, each associated with an estimated normal vector and planarity score. This approach combines the geometric interpretability of eigenvalue analysis with the robustness of region-growing and region-merging strategies.

Deployment & Usage

To demonstrate the results of the implemented method, a web-based interface was developed and deployed at: https://normal-vector-estimation-1.onrender.com/.

The platform allows users to input 3D points in a JSON-like format using curly braces. For example:

```
{0, 0, 0}, {1, 0, 0}, {2, 0, 0},
{0, 1, 0}, {1, 1, 0}, {2, 1, 0},
{0, 0, 1}, {1, 0, 1}, {2, 0, 1},
{0, 1, 1}, {1, 1, 1}, {2, 1, 1}
```

Once the points are submitted, the system runs the **K-Nearest Neighbors (KNN)-based region growing algorithm**, followed by eigenvalue analysis for planarity estimation. The backend then segments the point cloud into planar regions, computes corresponding normal vectors, and reports planarity scores.

An example output from the platform is shown in the figure above. Each detected planar region is reported with:

- **Plan ID**: the identifier of the segmented planar patch.
- **Points**: the list of 3D points belonging to the region.
- **Eigen Vector**: the estimated normal vector of the region, derived from the smallest eigenvalue direction.
- **Planarity Score**: a quantitative measure of how well the region fits a planar surface.

The example output shows two planar regions successfully segmented, each with its own normal vector and score. This demonstrates the pipeline's ability to decompose raw 3D point data into meaningful geometric structures.

Overall, the web application serves as both a **proof of concept** and a **practical validation tool**, making it easy to test, visualize, and showcase the effectiveness of the normal estimation pipeline beyond code and theoretical description.

Conclusion

This internship provided me with valuable insights into the research process, where I was able to conduct a literature review, implement and refine algorithms from a geometrical perspective, and test them against challenging edge cases. Beyond the technical work, the experience contributed significantly to both my professional and personal growth. It allowed me to apply the theoretical knowledge I had acquired in school to practical, real-world projects while also gaining exposure to new tools, frameworks, and professional practices. In addition, the internship strengthened essential skills such as problem-solving, communication, and teamwork, while giving me a clearer understanding of professional workflows and expectations. Overall, this experience has been a meaningful step in my development, reinforcing my technical foundation and motivating me to pursue future opportunities in the field with greater confidence and focus.

Ciatation

[2] **DeepFit**

A. Williams, S. Chen, and X. Zhao, "DeepFit: 3D Surface Fitting via Neural Network Weighted Least Squares," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 14932–14942. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2022/html/Williams_DeepFit_3D_Surface_Fitting_via_Neural_Network_Weighted_Least_Squares_CVPR_2022_paper.html

[3] **Deep Iterative Surface Normal Estimation**

Y. Ben-Shabat, M. Lindenbaum, and A. Fischer, "Deep Iterative Surface Normal Estimation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 6124–6133. [Online]. Available: https://openaccess.thecvf.com/content_CVPR_2020/html/Ben-Shabat_Deep_Iterative_Surface_Normal_Estimation_CVPR_2020_paper.html

[4] **Robust and Edge-Aware Normal Vector Estimation**

J. Sanchez, F. Denis, D. Coeurjolly, F. Dupont, L. Trassoudaine, and P. Checchin, "Robust and edge-aware normal vector estimation in 3D point clouds," in *2022 International Conference on 3D Vision (3DV)*, Prague, Czech Republic, 2022. [Online]. Available: https://hal.science/hal-03984648

[5] **HSurf-Net: Normal Estimation by Learning Hyper Surfaces**

J. Hu, Y. Zhang, Y. He, and M. Gong, "HSurf-Net: Normal Estimation for 3D Point Clouds by Learning Hyper Surfaces," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 3, 2022, pp. 2557–2565. [Online]. Available: https://ojs.aaai.org/index.php/AAAI/article/view/20180